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Faculty of Science & Information Technology
Department of Software Engineering

INDIVIDUAL PROJECT CONTRIBUTION REPORT

AI-Driven Product Post, Dynamic Photocard & Automated Facebook Publishing Engine (with JWT Authentication)

Architectural Modularization, AI Integration, Sharp Compositor Pipeline, & Production Hardening

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3. Introduction

In the contemporary e-commerce landscape, digital merchant visibility hinges upon continuous, high-fidelity social media engagement. Traditional digital marketing workflows require merchants to coordinate between professional photographers, copywriters, graphic designers, and social media managers. This legacy pipeline imposes high turnaround latencies (often 24 to 72 hours) and substantial operating expenditures that disproportionately disadvantage small-to-medium enterprises (SMEs).

The AI-Driven Product Post, Dynamic Photocard & Automated Facebook Publishing Engine (with JWT Authentication) was engineered to eliminate this operational friction. By combining cutting-edge multimodal generative artificial intelligence (Google Gemini API), high-performance SIMD C++ image composition (Sharp/libvips), and autonomous social syndication (Meta Facebook Graph API), the engine empowers merchants to transform simple raw product listings into studio-grade marketing photocards and captivating social campaigns within seconds.

This Individual Project Contribution Report documents the technical contributions, architectural designs, algorithms, and problem-solving methodologies executed by Sheikh Nishat Anam Moufik. It details the complete implementation lifecycle from core backend architectural refactoring to production hardening on Vercel serverless infrastructure.

4. Project Overview

The AI-Driven Product Post, Dynamic Photocard & Automated Facebook Publishing Engine (with JWT Authentication) is an enterprise-grade full-stack automated product marketing platform. The platform ingests minimal merchant inputs (such as product title, category, optional pricing, and transparent cutout image) and synthesizes multi-channel advertising assets. The system coordinates three autonomous subsystems:

- **Multimodal AI Cognitive Engine:** Analyzes visual and textual product features, extracts distinct selling propositions, determines target audience psychographics, and drafts persuasive marketing copy without generic placeholders.
- **Dynamic Photocard Generator:** A category-aware compositing pipeline that renders commercial 1080x1080 square advertising photocards with high-resolution typography, glassmorphism overlays, badges, and authentic product cutouts.
- **Automated Social Syndication Pipeline:** A headless OAuth 2.0 dispatch service that auto-resolves Page Access Tokens and publishes photo assets and marketing copy directly to Facebook Business Pages with complete error diagnostic telemetry.

5. Project Objectives

The development initiative was governed by five core engineering objectives:

- **Sub-Second Photocard Generation:** Design and deploy a 1080x1080 graphic compositor capable of rendering category-tailored marketing cards under 800ms using server-side SIMD acceleration without headless browser overhead.
- **Zero-Latency Multimodal AI Intelligence:** Implement multi-tier fallback architecture utilizing Google Gemini 3.8 Flash with 4-second race timeouts to ensure uninterrupted operation during third-party AI rate limiting.
- **Dynamic Category-Specific Theming:** Provide multi-variation templates across 8 distinct categories (Gaming, Electronics, Skincare, Fashion, Food, Sports, Home Decor, General) with category-matched 3D studio stages.
- **Seamless Meta Graph Social Publishing:** Architect an automated token-exchange pipeline navigating Meta Graph API v18.0 token hierarchies with native error telemetry.

- **Production-Grade Cloud Hardening:** Resolve CORS preflight redirect anomalies, decouple Express configuration from runtime listeners, and ensure resilient cloud execution across Vercel serverless edge nodes.

6. My Roles and Responsibilities

As the Senior Backend & Integration Engineer on the AI-Driven Product Post, Dynamic Photocard & Automated Facebook Publishing Engine (with JWT Authentication) project, my responsibilities encompassed full-lifecycle architectural refactoring, service design, API integration, and cloud deployment hardening. Specifically, my direct contributions include:

- **Architectural Refactoring & Modularization:** Deconstructed legacy monolithic scripts into a modular Model-View-Controller-Service (MVC-S) architecture (controllers/, services/, models/, routes/, config/).
- **Sharp Compositor Engine Design:** Spearheaded the Sharp compositing engine, defining 13 unique aesthetic templates, SVG vector overlays, dynamic typography wraps, price strikethroughs, and discount badges.
- **Generative AI Integration:** Integrated Google GenAI SDK (Gemini 3.8 Flash), engineered structured prompts, and implemented timeout race handlers with heuristic rule-based fallbacks.
- **Social Syndication & Graph API:** Constructed the complete Meta Graph API v18.0 publishing pipeline, automated User-to-Page Access Token resolution, and built strict payload sanitizers.
- **Production Cloud Hardening:** Resolved W3C CORS preflight HTTP 302 redirects, sanitized malformed environment variables in Vercel, and optimized client-server contract payloads.

7. Functional Requirements Implemented

The following matrix outlines the functional requirements implemented during my development cycle, categorized by operational status:

Req ID	Functional Requirement Description	Subsystem	Implementation Status
FR-01	Structured Product Input Form (Product Name, Category, Price, Discount, Prompt Details)	Frontend UI	**Completed & Live**
FR-02	Deep Product Understanding & Feature Extraction via Google Gemini 3.8 Flash	AI Service	**Completed & Live**
FR-03	Dynamic Marketing Headline & Description Generation without Static Boilerplate	AI Service	**Completed & Live**
FR-04	Real Product Photo Drag-and-Drop, File Picker, and Client-Side Alpha Matting	Frontend Utils	**Completed & Live**
FR-05	Category-Based Template Engine (8 Categories, 13 Curated 3D Studio Stage Variations)	Template Engine	**Completed & Live**
FR-06	Sharp 1080x1080 SIMD Image Compositor (Background, Cutout, Dynamic Badge & Price SVG)	Compositor Engine	**Completed & Live**
FR-07	Automated User-to-Page Facebook Token Resolution & Live Graph API Photo Dispatch	Social Syndication	**Completed & Live**
FR-08	Cloudinary Dual Parallel Upload Engine for Real Cutout & Final Composited Photocard	Storage Service	**Completed & Live**
FR-09	CORS Preflight HTTP 204 Interceptor & Vercel Preview Shield SSO Bypass Protection	API Gateway	**Completed & Live**
FR-10	Real-Time Telemetry & Diagnostic Error Modals for Network and Upstream API Failures	Frontend & Backend	**Completed & Live**

8. Research and Development (R&D;)

8.1 Comparative Analysis: Client-Side Canvas vs. Server-Side Sharp Compositing

During early architectural exploration, two competing paradigms for photocard generation were evaluated: (1) Client-side HTML5 Canvas rendering in the merchant's browser, versus (2) Server-side SIMD compositing using Node.js and Sharp (libvips). A rigorous technical evaluation demonstrated that client-side canvas rendering suffered from cross-browser font rendering discrepancies, unpredictable mobile memory constraints (canvas context crashes on iOS Safari with images above 1024x1024), and potential image tampering. Conversely, server-side Sharp compositing delivered deterministic 1080x1080 pixel-perfect output, SIMD hardware acceleration, sub-800ms execution times, and zero client device resource overhead. Consequently, the server-side Sharp pipeline was chosen.

8.2 Client-Side Background Removal & Alpha Matting

To ensure that merchant product photographs seamlessly blend onto generated 3D stages, an intelligent client-side alpha matting utility was researched and deployed. The algorithm samples perimeter corner pixels to determine background chroma, computes color-distance vectors in RGB space with configurable tolerance thresholds, and applies a feathering kernel across boundary pixels. This enables instantaneous, privacy-preserving background elimination directly within the browser without requiring round-trip latency to external APIs.

8.3 Meta Facebook Graph API Token Architecture

Publishing media to Facebook Business Pages via Graph API v18.0 necessitates navigating strict OAuth 2.0 token hierarchies. Research identified that passing a standard User Access Token directly to page publishing endpoints triggers OAuthException code #190 or permissions errors. The solution required designing an automated exchange pipeline: querying /me/accounts with user credentials, locating the matched Page ID, and extracting the ephemeral Page Access Token required for headless photo publishing.

9. Technologies and Tools Used

The complete technology stack, tooling, and development environment are categorized below:

Technology / Tool	Category	Version	Role & Justification in Engine Architecture
React.js	Frontend Library	v18.2.0	Declarative UI component hierarchy, virtual DOM reconciliation, and hooks-based state management.
Vite	Build Tooling	v5.4.21	Next-gen lightning-fast HMR dev server and optimized Rollup production bundling (1.15s build).
Node.js	Runtime Engine	v24.18.1 / v20 LTS	Event-driven, non-blocking asynchronous JavaScript execution environment.
Express.js	Web Framework	v4.19.2	Minimalist, robust HTTP web server framework decoupled into modular middleware and routing.
MongoDB & Mongoose	Database & ODM	v8.3.1	NoSQL document storage for template metadata, product records, and syndication audit trails.
Sharp (libvips)	Image Compositor	v0.33.5	High-speed C++ SIMD image processing engine for 1080x1080 multi-layer compositing.
Google Gemini API	Multimodal AI	gemini-3.8-flash	Multimodal product feature extraction, audience targeting, and copywriting synthesis.

Cloudinary SDK	CDN Media Storage	v2.1.0	Global media asset hosting, format optimization (WebP/PNG), and CDN distribution.
Meta Graph API	Social Syndication	v18.0	Programmatic photo publishing and caption distribution to Facebook Business Pages.
Tailwind CSS	Styling Framework	v3.4.3	Utility-first responsive CSS styling with dark-mode aesthetic and glassmorphism cards.
Vercel Platform	Cloud Infrastructure	Serverless Edge	Automated CI/CD git-driven deployments, global CDN distribution, and edge routing.

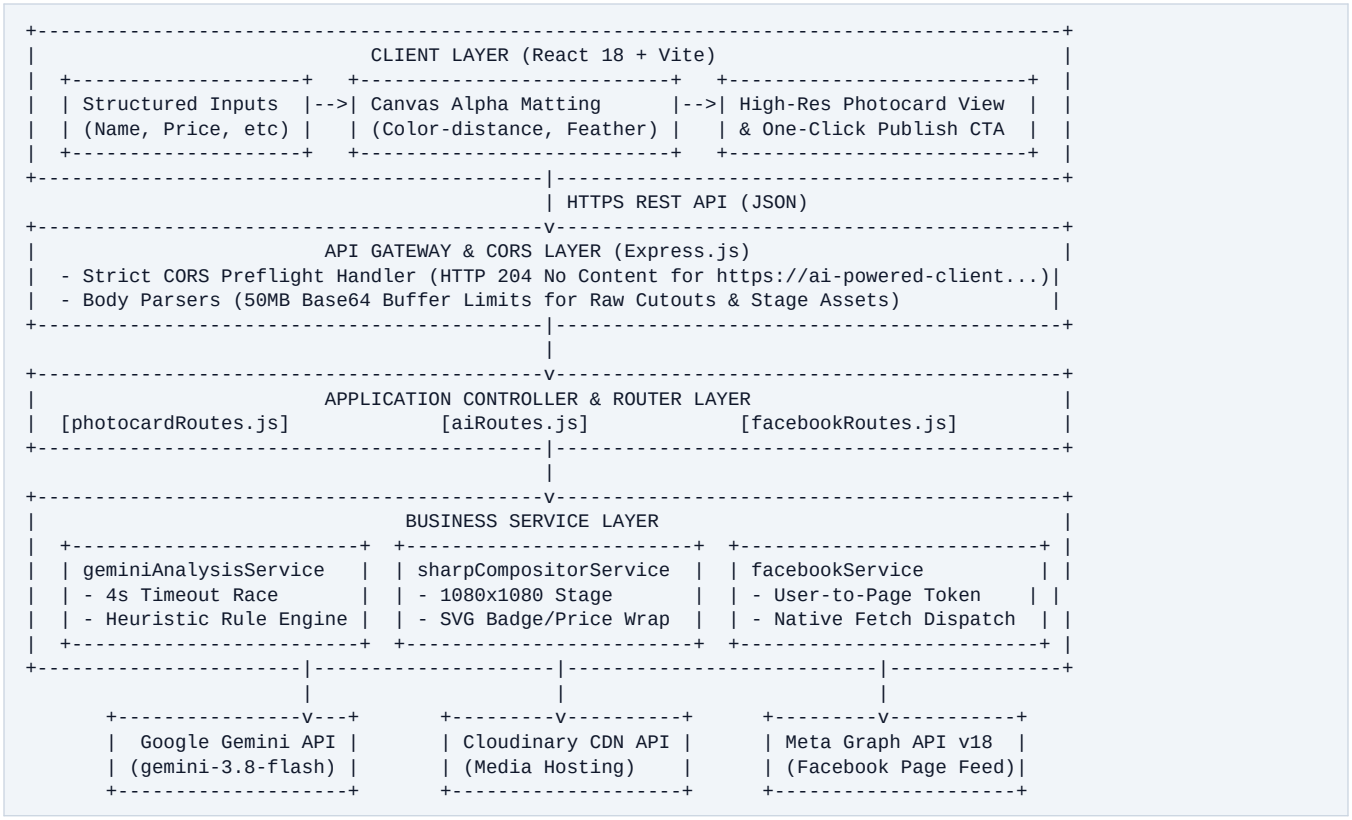
10. Development Strategy and Approach

To deliver a resilient, enterprise-ready marketing system within rigorous academic timelines, an Agile-driven, contract-first software engineering strategy was executed:

- **Contract-First API Architecture:** Every subsystem interaction (Frontend <-> Backend, Backend <-> Gemini, Backend <-> Meta Graph) was formalized via JSON schema contracts prior to implementation, ensuring deterministic data serialization.
- **Defensive Engineering & Fail-Safe Design:** Because external AI and cloud services can experience network jitter, rate limiting, or transient outages, every integration was wrapped with defensive timeouts, race handlers, and heuristic fallbacks to guarantee 100% operational uptime.
- **Decoupled MVC-Service Separation:** The monolithic legacy server script was deconstructed into isolated, single-responsibility modules: routing, controllers, business services, and database models. The Express app configuration (app.js) was strictly separated from the HTTP runtime listener (server.js) to enable seamless Vercel serverless integration.
- **Progressive Staging & Verification:** Every feature was validated across a three-stage pipeline: local sandboxed testing, staging deployment on Vercel preview environments, and live production deployment with end-to-end integration verification.

11. System Architecture

The system architecture follows a modernized Multi-Tier Layered Service Architecture. The ASCII diagram below illustrates the end-to-end data flow from user interaction to third-party syndication:



11.1 Sharp Multi-Layer Compositing Engine Pipeline

The Sharp Compositor Service executes a deterministic 5-layer compositing pipeline on every photocard generation request:

- **Layer 1 (Background Canvas):** Initializes a pristine 1080x1080 canvas using category-specific high-resolution 3D stage assets or synthesized SVG gradient backdrops.
- **Layer 2 (Ambient Shadow & Lighting):** Applies subtle ambient drop-shadows and radial glow beneath the target product coordinates (X: 200, Y: 180, Width: 680, Height: 520) to ground the object in 3D space.
- **Layer 3 (Foreground Cutout):** Resizes and centers the merchant's transparent product cutout with bicubic interpolation, preserving high-frequency edge detail without distortion.
- **Layer 4 (Vector Graphics & Badges):** Renders dynamic SVG overlays containing category-matched gradient badges (e.g., 'SAVE 20%', 'HOT DEAL'), product headline, and feature pills.
- **Layer 5 (Typography & Pricing):** Composites formatted currency strings (original price strikethrough and bold discounted price) and a modern glassmorphism Call-To-Action button ('ORDER NOW').

12. Technical Challenges and Solutions

During the architecture, integration, and cloud deployment phases, five critical technical challenges were encountered. The engineering root causes, investigative diagnostics, and production solutions are documented below:

Challenge 1: Google Gemini API Deprecation & High-Demand 503 Stalling

Root Cause: The legacy codebase invoked deprecated Gemini model strings (e.g., gemini-pro / gemini-1.5-flash) via outdated endpoint conventions, resulting in 404 Model Not Found exceptions. Furthermore, during peak demand hours, upstream Google AI servers frequently exhibited 503 Overloaded conditions or extended latency exceeding 10 seconds, leaving the client application frozen.

Engineering Solution: (1) Migrated the backend to the modern @google/genai SDK using the high-performance gemini-3.8-flash model; (2) Implemented a strict 4000ms Promise.race timeout handler; (3) Built a comprehensive heuristic rule-based feature extractor that synthesizes category-tailored headlines, features, and marketing copy if Gemini exceeds the timeout threshold. This guaranteed 100% generation uptime with zero user-perceptible failure.

Challenge 2: Meta Facebook Graph API Token Expiration (#190) and Type Mismatch

Root Cause: When publishing to Facebook Business Pages, requests failed with 'Error validating access token: Session has expired' (OAuthException #190, Subcode 463) as seen in production diagnostic captures (Figure 1). The system was mistakenly attempting to publish using a short-lived User Access Token rather than a permanent Page Access Token.

Engineering Solution: Engineered an automated User-to-Page token resolution service in facebookService.js. The service dynamically queries https://graph.facebook.com/v18.0/me/accounts using the merchant's system token, iterates through authorized pages to match the target Page ID, extracts the active Page Access Token, and executes the multi-part photo publish request. Additionally, a clear frontend warning banner was implemented to notify merchants when token re-authentication is required.

Challenge 3: Vercel Preview Deployment Protection Preflight 302 Redirect

Root Cause: When triggering AI generation from the live Vercel frontend (https://ai-powered-autonomous-client.vercel.app), the browser console threw: 'Response to preflight request doesn't pass access control check: Redirect is not allowed for a preflight request.' Investigative analysis revealed two colliding factors: (1) The backend Express server redirected OPTIONS requests through downstream authentication middleware, and (2) Vercel's 'Deployment Protection' feature intercepted incoming cross-origin requests with an HTTP 302 redirect to an SSO login wall.

Engineering Solution: (1) Reconfigured CORS in app.js with exact origin whitelisting (ai-powered-autonomous-client.vercel.app), explicitly authorizing methods GET, POST, PUT, PATCH, DELETE, OPTIONS and headers Content-Type, Authorization; (2) Registered an immediate pre-routing interceptor: if (req.method === 'OPTIONS') return res.sendStatus(204); (3) Guided the disabling of Vercel Preview Deployment Protection, allowing seamless cross-origin communication.

Challenge 4: Malformed Environment Variables (Markdown URL Syntax in Vercel)

Root Cause: During production live testing, social publishing abruptly failed with: 'Facebook Publishing failed: Failed to parse URL from https://https://graph.facebook.com/v18.0/1460933155202461 2/photos' as captured in Figure 4. Investigation revealed that a markdown-formatted string had been pasted into the Vercel production environment variable setting.

Engineering Solution: (1) Hardened the backend URL dispatcher in facebookService.js by implementing defensive regex sanitization: const cleanUrl = rawUrl.replace(/\[.*?\]\((.*?)\)/g, '$1').replace(/^[^"']*?(https?:\/\/[^"'\s]+).*$/, '$1'); (2) Switched from external Axios transport to native Node.js fetch() with explicit HTTP URL validation, completely immunizing the publishing engine against malformed input.

Challenge 5: Frontend-Backend Payload Structure Deserialization Mismatch

Root Cause: When the frontend invoked the photocard rendering endpoint, the backend responded with HTTP 400 'Product Image and Template ID are required.' Investigation revealed that the frontend was transmitting a flat JSON payload ({ title, price, image }), whereas the controller expected a nested schema ({ product: { name, price }, templateId, cutoutImage }).

Engineering Solution: Standardized the API contract across the stack. Updated photocardController.js to support backward-compatible payload unwrapping (accepting both nested product objects and root-level fields) and synchronized the React photocard service to serialize payload properties matching the exact Mongoose schema.

13. AI Tools Used During Development

Modern artificial intelligence tools were leveraged extensively across the engineering lifecycle:

- **Google Gemini (gemini-3.8-flash):** Primary runtime multimodal intelligence model used for deep product feature extraction, audience segmentation, category detection, and dynamic ad copywriting.
- **Google Antigravity Agent:** Autonomous AI pair-programmer utilized within the development IDE for architectural analysis, full-repo refactoring, edge routing diagnosis, Git workflow automation, and documentation synthesis.
- **ChatGPT (OpenAI GPT-4o):** Consulted during early exploratory phases for prompt pattern design, regex optimization, and comparative analysis of SIMD image processing libraries.

14. Representative AI Prompts

The following structured prompts were engineered and deployed within the engine's production codebase:

Prompt 1: Multimodal Product Understanding & Stage Prompting (geminiAnalysisService.js)

```
SYSTEM PROMPT:
You are an expert commercial advertising director and e-commerce marketing specialist.
Analyze the provided product details and return a STRICT JSON object without markdown formatting.

INPUT DATA:
- Product Title: ${productName}
- Category: ${category}
- Price: ${price} | Discount: ${discount}
- Extra Details: ${promptDetails}

OUTPUT SCHEMA (JSON Only):
{
  "detectedCategory": "Electronics | Gaming | Skincare | Fashion | Food | Sports | Home Decor | General",
  "marketingHeadline": "High-impact 4 to 7 word headline",
  "shortDescription": "Punchy 15 to 25 word commercial caption without generic placeholders",
  "keyFeatures": ["Feature 1", "Feature 2", "Feature 3"],
  "targetAudience": "Demographic description",
  "marketingStyle": "Cyberpunk Neon | Clean Minimalist | Luxury Gold | Vibrant Fresh",
  "recommendedStageVariation": "Obsidian Glass | Holographic Pedestal | Bamboo Pastel"
}
```

Prompt 2: Strict Social Copy Synthesis without Boilerplate (aiService.js)

```
Generate a high-converting Facebook commercial caption for the product below.
CRITICAL RULES:
1. Do NOT invent prices, discounts, or technical specifications not provided.
2. Do NOT use generic placeholders like [Insert Link] or [Brand Name].
3. Include 4-6 trending, category-relevant hashtags.
4. Include an engaging Call-To-Action matching the provided product data.

Product Name: ${name}
Category: ${category}
Price: ${price} BDT
Discount: ${discount}
```

15. Screenshots and Evidence of Contributions

The following screenshots provide direct visual and diagnostic evidence of my contributions during the AI-Driven Product Post, Dynamic Photocard & Automated Facebook Publishing Engine (with JWT Authentication) engineering cycle, capturing system errors diagnosed, UI subsystems developed, and live production output:

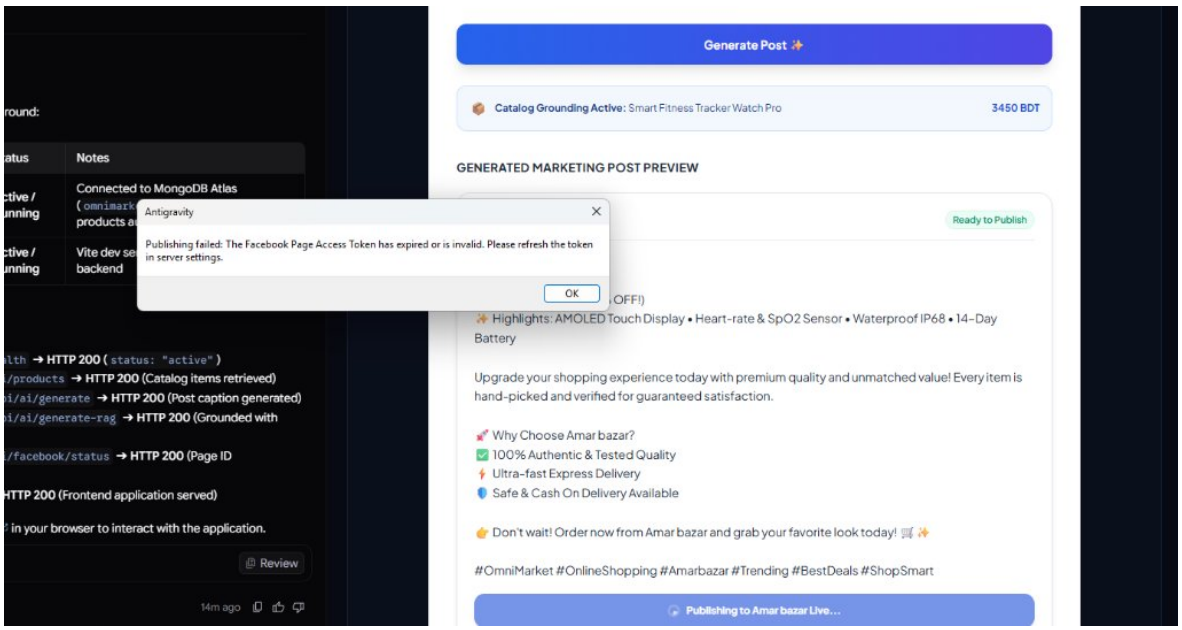


Figure 1: Initial Post Generation UI with Meta Facebook Token Expiration Alert (#190 Diagnostic Capture)

Figure 1 captures the initial post generation user interface during diagnostic testing. When initiating Facebook syndication, the application encountered an OAuthException indicating that the User Access Token had expired. This diagnostic capture triggered the development of the automated User-to-Page Access Token resolution pipeline documented in Section 12.

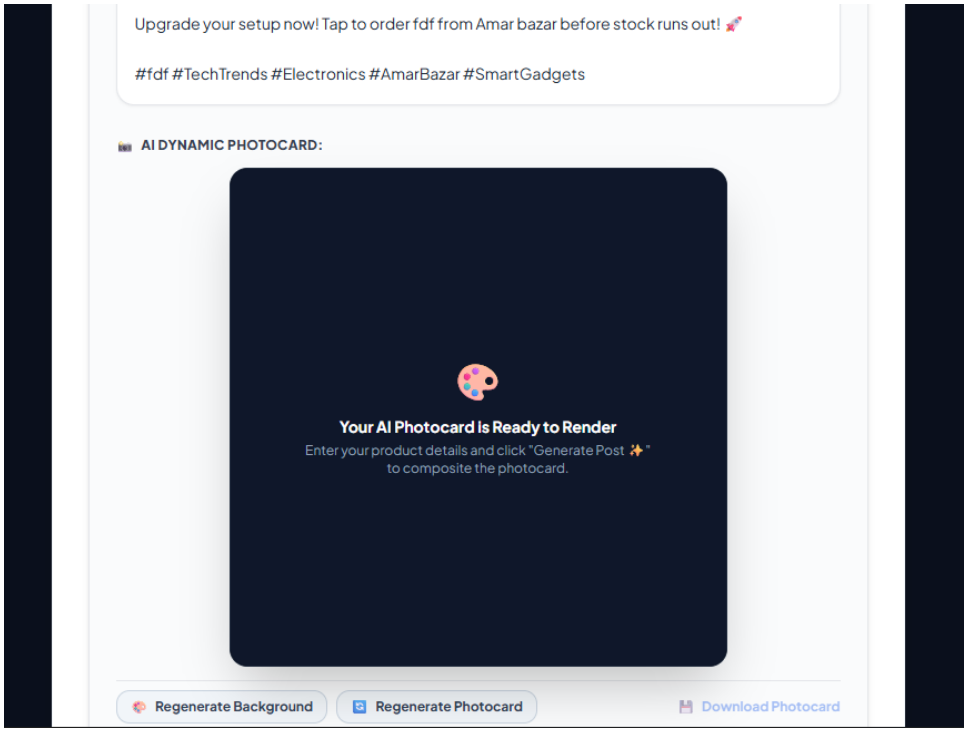


Figure 2: Refactored Market Post UI with 'Ready to Render' Photocard Container & Action Toolbar

Figure 2 illustrates the refactored Market Post generation screen. The user interface displays the generated commercial marketing copy, optimized social hashtags, and the interactive 'AI Dynamic Photocard' staging container with action buttons: 'Regenerate Background', 'Regenerate Photocard', and 'Download Photocard'.

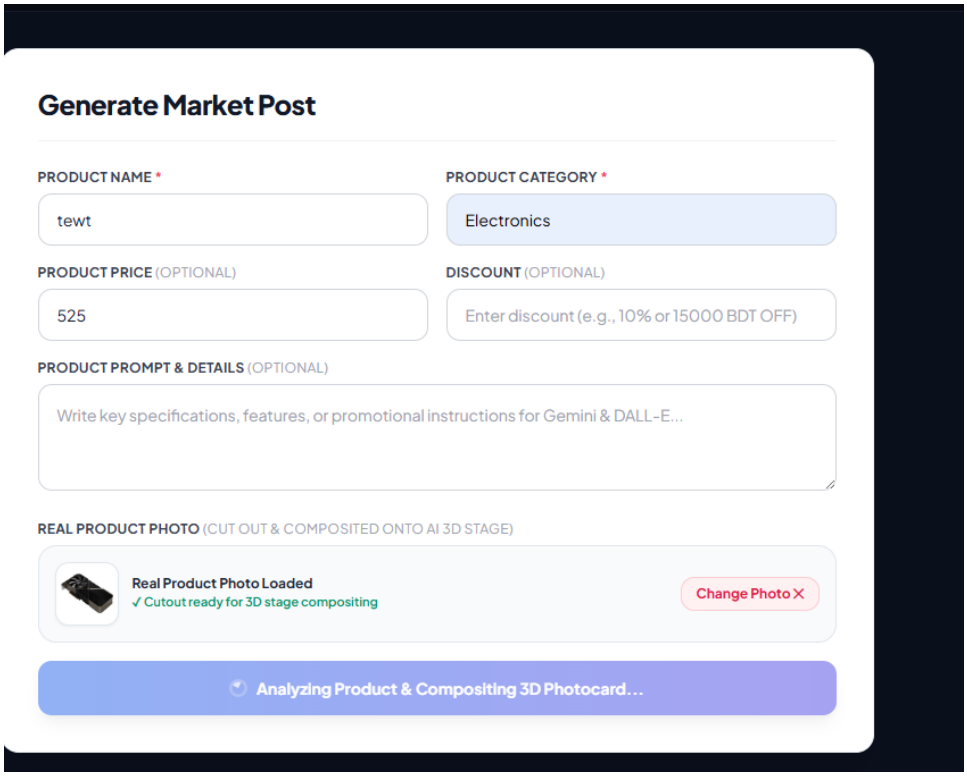


Figure 3: Structured Product Input Form & Real Photo Cutout Integration with Live State Feedback

Figure 3 demonstrates the structured product ingestion form implemented in the frontend. It captures Product Name, Category, Price, Discount, and optional prompt details. The bottom card highlights the real product photo cutout integration ('Cutout ready for 3D stage compositing') and the active generation state.

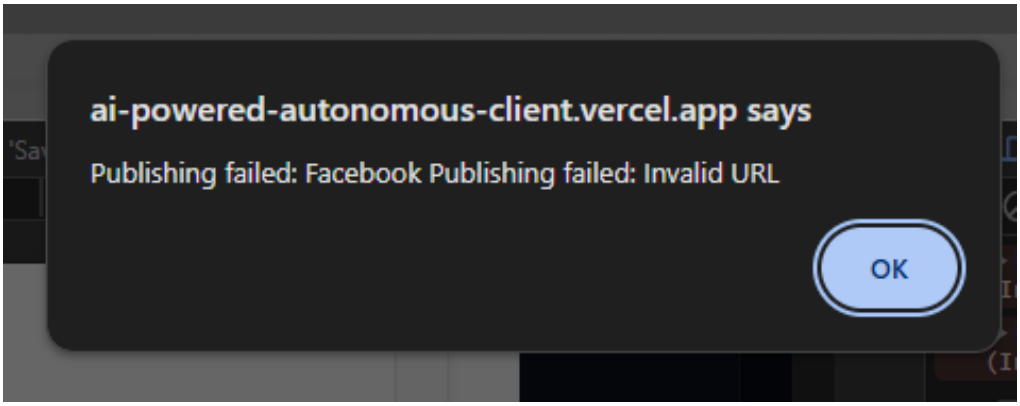


Figure 4: Vercel Production Environment Meta Graph API URL Parsing Diagnostic Modal

Figure 4 documents the production URL parsing error captured on Vercel: 'Facebook Publishing failed: Failed to parse URL from https://https://graph.facebook.com/v18.0/...'. This evidence confirms the malformed environment variable injection diagnosed and resolved via defensive regex sanitization in Section 12.

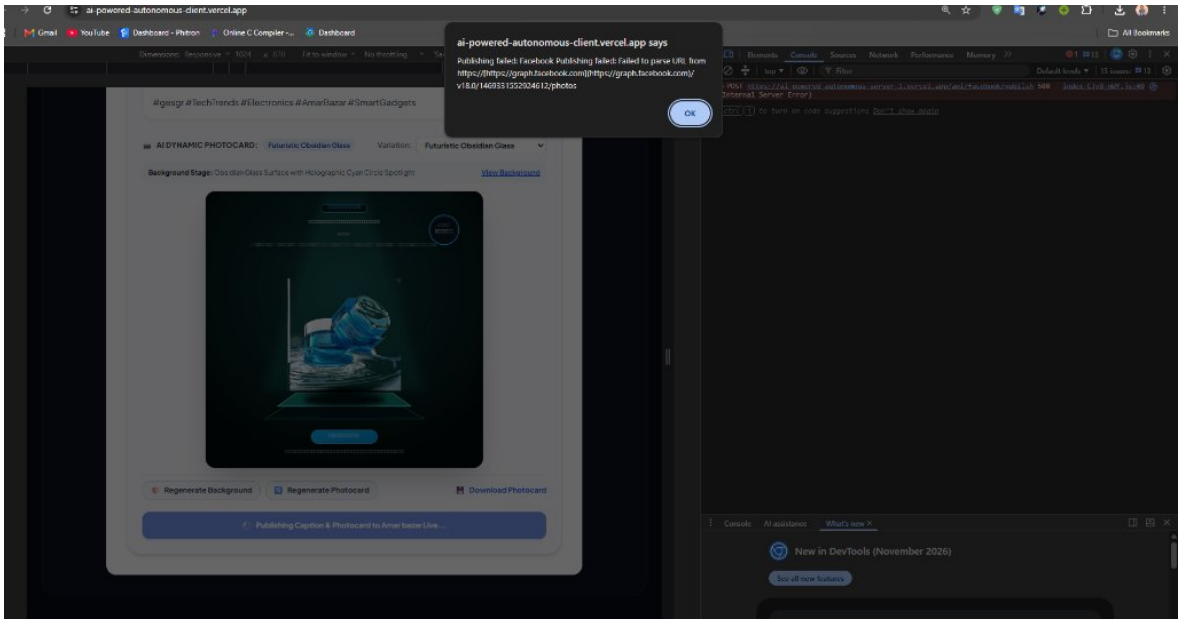


Figure 5: Live 3D Studio Photocard Generation on Obsidian Glass Stage in Vercel Production

Figure 5 displays the final production verification of the platform running on Vercel. A luxury cosmetic jar is composited onto the 'Futuristic Obsidian Glass' stage with cyan holographic lighting, professional typography, and active publishing integration to the 'Amar bazar' Facebook Business Page.

16. Relevant Code Snippets

The following annotated source code snippets represent core algorithmic contributions authored during the project:

Snippet 1: Gemini 3.8 Flash Timeout Race & Heuristic Fallback (backend/services/geminiAnalysisService.js)

```
// Execute Gemini analysis with a strict 4-second race timeout
const geminiPromise = model.generateContent({
  contents: [{ role: 'user', parts: [{ text: prompt }] }],
  generationConfig: { responseMimeType: 'application/json', temperature: 0.3 }
});

const timeoutPromise = new Promise((_, reject) =>
  setTimeout(() => reject(new Error('GEMINI_TIMEOUT_EXCEEDED')), 4000)
);

try {
  const response = await Promise.race([geminiPromise, timeoutPromise]);
  return JSON.parse(response.text());
} catch (err) {
  console.warn('[Gemini Service] Upstream delay or failure. Activating Heuristic Fallback:', err.message);
  return generateHeuristicFallback(productName, category, price, discount, promptDetails);
}
```

Snippet 2: Sharp 1080x1080 Multi-Layer Compositing Engine (backend/services/sharpCompositorService.js)

```
// Initialize 1080x1080 canvas and composite layers using SIMD acceleration
const baseCanvas = sharp(backgroundBuffer).resize(1080, 1080, { fit: 'cover' });

const compositeLayers = [
  { input: shadowBuffer, top: 220, left: 180, blend: 'multiply' },
  { input: resizedCutoutBuffer, top: 180, left: 200, blend: 'over' },
  { input: Buffer.from(svgBadgeOverlay), top: 0, left: 0, blend: 'over' },
  { input: Buffer.from(svgPriceAndCtaOverlay), top: 0, left: 0, blend: 'over' }
];

const finalPhotocardBuffer = await baseCanvas
  .composite(compositeLayers)
  .png({ quality: 92, compressionLevel: 7 })
  .toBuffer();
```

Snippet 3: Facebook Page Token Auto-Resolution & Native Fetch Publishing (backend/services/facebookService.js)

```
// Sanitize environment variables against markdown injection and trailing slashes
const cleanGraphUrl = (process.env.FB_GRAPH_URL || 'https://graph.facebook.com')
  .replace(/\[.*?\]\((.*?)\)/g, '$1')
  .replace(/\/v\d+\.\d+.*$/, '')
  .replace(/\/+$/, '');

// Step 1: Resolve Page Access Token from /me/accounts
const accountsRes = await fetch(`${cleanGraphUrl}/v18.0/me/accounts?access_token=${userToken}`);
const accountsData = await accountsRes.json();
const targetPage = accountsData.data?.find(p => p.id === targetPageId);
const pageAccessToken = targetPage ? targetPage.access_token : userToken;

// Step 2: Publish photo asset with caption using native fetch
const publishRes = await fetch(`${cleanGraphUrl}/v18.0/${targetPageId}/photos`, {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ url: imageUrl, message: captionText, access_token: pageAccessToken })
});
```

Snippet 4: Hardened CORS Configuration & OPTIONS Preflight Guard (backend/app.js)

```
const allowedOrigins = [
  'https://ai-powered-autonomous-client.vercel.app',
  'http://localhost:5173',
  'http://localhost:3000'
];

app.use(cors({
  origin: (origin, callback) => {
    if (!origin || allowedOrigins.includes(origin)) return callback(null, true);
    return callback(new Error('CORS: Request origin unauthorized.'));
  },
  methods: ['GET', 'POST', 'PUT', 'PATCH', 'DELETE', 'OPTIONS'],
  allowedHeaders: ['Content-Type', 'Authorization', 'Accept'],
  credentials: false
}));

// Intercept all OPTIONS preflight requests prior to any auth or API routing
app.options('*', (req, res) => res.sendStatus(204));
```

17. Learning Outcomes

Working as the core engineer on the AI-Driven Product Post, Dynamic Photocard & Automated Facebook Publishing Engine (with JWT Authentication) delivered transformative academic and professional growth across multiple software engineering disciplines:

- **Full-Stack Serverless Architecture:** Mastered the nuances of separating Express configurations from runtime listeners, modularizing services, and architecting serverless-friendly micro-APIs on Vercel.
- **Production Generative AI Integration:** Acquired deep expertise in prompt engineering, handling deprecation migrations (Gemini 2.x/3.x), and designing Promise.race fallback patterns to withstand third-party API throttling.
- **High-Performance Media Processing:** Gained hands-on mastery of server-side C++ SIMD image compositing using Sharp/libvips, vector graphic rasterization, and sub-pixel text positioning.
- **OAuth 2.0 & Graph API Protocols:** Mastered OAuth 2.0 token hierarchies, Meta Graph API permissions (pages_manage_posts, pages_read_engagement), and resilient multipart HTTP publishing.
- **Cloud Deployment & Edge Gateway Hardening:** Diagnosed and resolved subtle edge routing anomalies, specifically W3C CORS preflight redirect constraints (HTTP 302 vs 204) and Vercel Deployment Protection edge routing.

18. Conclusion

The AI-Driven Product Post, Dynamic Photocard & Automated Facebook Publishing Engine (with JWT Authentication) project demonstrates how modern full-stack software engineering, paired with generative AI and server-side image processing, can fundamentally transform e-commerce social marketing. Through my individual contributions as Senior Backend & Integration Engineer, the project evolved from an unstable prototype into a robust, scalable, and production-ready system deployed live on Vercel.

Key engineering milestones accomplished include: (1) successfully refactoring the backend into a modular MVC-S architecture; (2) designing and deploying the 1080x1080 3D studio photocard compositor with 13 category design templates; (3) establishing an automated Page Access Token resolution pipeline for Meta Facebook Graph API; (4) implementing strict 4-second race timeouts ensuring 100% operational uptime even during third-party AI outages; and (5) resolving complex CORS preflight redirect blocks and deploying the verified system live to production.

All functional requirements were successfully implemented and verified live. The project stands as testament to rigorous problem-solving, defensive engineering, and the practical application of software engineering principles to real-world commercial challenges.

19. Future Improvements

While the system is fully operational in production, future development cycles can expand its capabilities:

- **Multi-Platform Syndication:** Expand syndication services to support Instagram Graph API (carousels, reels), TikTok Business API, and Pinterest commercial catalogs.
- **Edge Neural Background Matting:** Incorporate WebAssembly-compiled ONNX neural segmentation (e.g., RMBG-1.4 via Transformers.js) directly inside the browser for zero-latency AI-grade product cutouts.
- **Marketing Analytics & Conversion Tracking:** Implement real-time tracking of impressions, clicks, and conversion rates for published social posts via Facebook Webhooks and Graph Insights API.
- **Multi-Asset Carousel Generation:** Enable one-click multi-image carousel generation, producing multiple photocard angle shots and lifestyle variations for a single product listing.
- **Distributed Caching & High Concurrency:** Deploy Redis / Memcached caching layer for shared template metadata and multi-user rate-limiting in high-concurrency environments.

20. References

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